

# The cyclic-portion subnormality question was answered negatively

Literature-status packet for arXiv:0906.5412

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**Status.** Literature already answered: full negative answer.

## Source question

Szafraniec assumes that the densely defined operator  $S$  has invariant domain,  $SD(S) \subseteq \mathcal{D}(S)$ . For  $f \in \mathcal{D}(S)$ , its cyclic portion is the restriction

$$S_f = S|_{\mathcal{D}_f}, \quad \mathcal{D}_f = \{p(S)f : p \in \mathbb{C}[Z]\},$$

viewed in the closure of  $\mathcal{D}_f$ . On source PDF page 21 he asks:

If every  $S_f$ ,  $f \in \mathcal{D}(S)$ , is subnormal, must  $S$  be subnormal?

The paper records affirmative results for bounded operators and several additional unbounded hypotheses, but leaves the general question open.

## Exact later answer

The answer is no. Szafraniec’s own abstract for his GPOTS 2012 talk, “*Subnormality and cyclicity*” in the negative, states on abstract-book PDF page 14 that the question whether subnormality of every cyclic portion forces subnormality of the operator was unresolved in the unbounded case and that the recent paper by Z. J. Jabłoński, I. B. Jung, and J. Stochel, *A non-hyponormal operator generating Stieltjes moment sequences*, JFA 262 (2012), 3946–3980, creates a counterexample.

The cited paper is arXiv:1110.0249. Its decisive Example 4.2.1, on supporting PDF page 21, constructs a weighted shift  $S_\lambda$  on a directed tree with one branching vertex. The operator generates Stieltjes moment sequences, yet is not hyponormal. Since every subnormal operator is hyponormal, it is not subnormal. The GPOTS abstract supplies the explicit authorial identification that this construction answers the cyclic-portion question itself, so the relation is not merely inferred from similar terminology.

## Scope and provenance

This packet records a literature result; it does not claim a new counterexample or reproduce the construction. The source question is exactly the unbounded invariant-domain local-to-global problem on cyclic portions. The supporting paper gives the counterexample, and the later abstract by the source author explicitly identifies it as a negative resolution of that question.

## References

1. F. H. Szafraniec, *Normals, subnormals and an open question*, Operators and Matrices 4 (2010), 485–510; arXiv:0906.5412.
2. Z. J. Jabłoński, I. B. Jung, and J. Stochel, *A non-hyponormal operator generating Stieltjes moment sequences*, J. Funct. Anal. 262 (2012), 3946–3980; arXiv:1110.0249.
3. F. H. Szafraniec, “*Subnormality and cyclicity*” in the negative, GPOTS 2012 Talks abstract book, p. 14.